

Capstone Finalized Visuals

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Bring in Cleaned & Transformed Dataset

```
Dataset <- read.csv("Cleaned_Transformed_Dataset_Klepps_V2.csv")
```

Import Needed Libraries

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(tidyverse)
library(scales)
library(stringr)
library(tidytext)
library(patchwork)
library(rlang)
library(ggrepel)
```

Perform Additional Cleaning

Removes all variations of the same title and groups based on a Family Title. Additionally nests into families with the ability to pull specifics within a family.

```
# Change case on common words to lower for traditional naming conventions
book_title_case <- function(x) {
  x |>
    str_to_title() |>
    str_replace_all(
      "\\b(Of|The|And|A|An|In|On|At|To|For|By|With|From|Or|But|Nor)\\b",
      ~ str_to_lower(.x)
    ) |>
    str_replace("^([a-z])", ~ str_to_upper(.x))
}

# Replace common characters; nest different titles & create book families (series)
```

```

Dataset_titles_nested <- Dataset |>
mutate(
  revenue = final_price * reviews_count,

  title_core = title |>
    str_to_lower() |>
    str_replace_all("'", "") |>
    str_replace_all(".*$", "") |>
    str_replace_all("\\(..*?\\)", "") |>
    str_replace_all("\\ba [a-z ]+ novel\\b", "") |>
    str_replace_all("\\ban [a-z ]+ novel\\b", "") |>
    str_replace_all("\\bbook\\s*[0-9]+\\b", "") |>
    str_replace_all("\\bbooks?\\s*[0-9]+\\b", "") |>
    str_replace_all("\\bvolumes?\\s*[0-9]+\\b", "") |>
    str_replace_all("\\bvol\\.?.*\\s*[0-9]+\\b", "") |>
    str_replace_all("\\bpart\\s*(one|two|three|four|five|six|seven|eight|nine|ten|[0-9]+)\\b", "") |>
    str_replace_all("\\bone\\s+and\\s+two\\b", "") |>
    str_replace_all("\\b(collector['']?s? edition|archival edition|archive edition|deluxe edition|special edition|illustrated edition|limited edition|anniversary edition|movie tie in edition|media tie in edition)\\b", "") |>
    str_replace_all("\\b(hardcover|paperback|mass market|kindle|audiobook|audio book|boxed set|box set|complete collection|collection|set|edition|deluxe|illustrated|special)\\b", "") |>
    str_replace_all("[[:punct:]]", " ") |>
    str_squish() |>
    book_title_case(),

  title_family = case_when(
    str_detect(str_to_lower(title_core),
      "harry potter|sorcerer's stone|chamber of secrets|prisoner of azkaban|goblet of fire|order of the phoenix|half blood prince|deathly hallows") ~
      "Harry Potter",

    str_detect(str_to_lower(title_core),
      "hunger games|catching fire|mockingjay|ballad of songbirds and snakes") ~
      "The Hunger Games",

    str_detect(str_to_lower(title_core),
      "fellowship of the ring|two towers|return of the king|the hobbit|lord of the rings") ~
      "Lord of the Rings / The Hobbit",

    str_detect(str_to_lower(title_core),
      "game of thrones|clash of kings|storm of swords|feast for crows|dance with dragons|song of ice and fire") ~
      "Game of Thrones",

    str_detect(str_to_lower(title_core),
      "divergent|insurgent|allegiant") ~
      "Divergent",

    str_detect(str_to_lower(title_core),

```

```

        "after|after we collided|after we fell") ~
    "After",

    str_detect(str_to_lower(title_core),
        "fifty shades of grey|fifty shades darker|fifty shades of freed|grey
        |freed|darker") ~
    "Fifty Shades of Grey",

    str_detect(str_to_lower(title_core),
        "killing lincoln|killing kennedy|killing jesus|killing patton|killing
        reagan|killing the rising sun|killing england|killing the ss|killing
        crazy horse|killing the mob|killing the
        killers|killing the legends|killing the witches") ~
    "Killing",

    str_detect(str_to_lower(title_core),
        "the way of kings|words of radiance|oathbringer|rhythm of war|wind
        and truth|edgedancer|dawnshard") ~
    "The Stormlight Archive",

    str_detect(str_to_lower(title_core),
        "one good deed|a gambling man|dream town") ~
    "Aloysius Archer",

    str_detect(str_to_lower(title_core),
        "long road to mercy|a minute to midnight|daylight") ~
    "Atlee Pine",

    str_detect(str_to_lower(title_core),
        "memory man|the last mile|the fix|the fallen|redemption|walk the
        wire|long shadows") ~
    "Amos Decker",

    str_detect(str_to_lower(title_core),
        "zero day|the forgotten|the escape|no man's land|daylight") ~
    "John Puller",

    str_detect(str_to_lower(title_core),
        "outlander|dragonfly in amber|voyager|drums of autumn|the fiery
        cross|a breath of snow and ashes|an echo in the bone|written in my
        own heart's blood|go tell the bees that i am gone") ~
    "Outlander",

    str_detect(str_to_lower(title_core),
        "from blood and ash| the crown of gilded bones|a kingdom of flesh
        and fire") ~
    "Blood and Ash",

    str_detect(str_to_lower(title_core),
        "shadow and bone|siege and storm|ruin and rising|six of crows|crooked
        kingdom|king of scars|rule of wolves") ~
    "Shadow and Bone",

```

```

    str_detect(str_to_lower(title_core),
               "the lost hero|the son of neptune|the mark of athena|the house of
               hades|the blood of olympus") ~
    "The Heroes of Olympus",

    str_detect(str_to_lower(title_core),
               "a court of thornes and roses|a court of mist and fury|a court of
               wings and ruin|a court of frost and starlight|a court of silver flames") ~
    "ACOTAR",

    str_detect(str_to_lower(title_core),
               "throne of glass|crown of midnight|heir of fire|queen of shadows
               |empire of storms|tower of dawn|kingdom of ash|the assassin's blade") ~
    "Throne of Glass",

    str_detect(str_to_lower(title_core),
               "twilight|new moon|eclipse|breaking dawn|midnight sun") ~
    "Twilight Saga",

    TRUE ~ title_core
  )
)

Dataset <- Dataset_titles_nested

# Franchise / series revenue
family_revenue <- Dataset |>
  group_by(title_family) |>
  summarize(
    total_revenue = sum(revenue, na.rm = TRUE),
    total_reviews = sum(reviews_count, na.rm = TRUE),
    number_of_titles = n_distinct(title_core),
    .groups = "drop"
  ) |>
  arrange(desc(total_revenue))

```

Theme Creation

```

theme_custom <- function() {
  theme_bw(base_size = 11) +
  theme(

    # Background
    panel.background = element_rect(fill = "#FFFFFF", color = NA),
    plot.background = element_rect(fill = "#FFFFFF", color = NA),

    # Gridlines
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_line(color = "#D9D9D9", linewidth = 0.3),

```

```

# Titles
plot.title = element_text(
  size = 20,
  face = "bold",
  hjust = 0.5,
  color = '#1F1F1F',
  margin = margin(b = 10)
),
plot.subtitle = element_text(
  size = 14,
  hjust = 0.5,
  color = '#3A3A3A',
  margin = margin(b = 10)
),
plot.caption = element_text(
  size = 12,
  face = "italic",
  color = '#4D4D4D',
  hjust = 1
),
axis.title.y = element_text(
  size = 16,
  face = "bold",
  color = '#1F1F1F',
  margin = margin(b = 10)
),
axis.title.x = element_text(
  size = 16,
  face = "bold",
  color = '#1F1F1F',
  margin = margin(b = 15)
),

# Axis Text
axis.text.y = element_text(
  size = 14,
  face = "bold",
  color = "#2E2E2E",
  lineheight = 1.1
),
axis.text.x = element_text(
  size = 14,
  face = "bold",
  color = "#2E2E2E",
),

# Facet headers
strip.background = element_rect(fill = "#EAF1F8", color = "#B6C7D8"),
strip.text = element_text(
  size = 10,
  face = "bold",
  color = "#1F1F1F"
),

```

```

# Legend
legend.title = element_text(size = 12, face = "bold"),
legend.text = element_text(size = 10, face = "bold"),

#Margins
plot.margin = margin(12, 35, 12, 12)
)
}

print("Custom Theme Created")

```

```
## [1] "Custom Theme Created"
```

1. Highest Grossing Genres

```

# Create revenue variable
Dataset <- Dataset |>
  mutate(revenue = final_price * reviews_count)

# Calculate total revenue per genre & include only top 5
genre_revenue_top5 <- Dataset |>
  group_by(genre) |>
  summarize(
    total_revenue = sum(revenue, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(total_revenue)) |>
  slice_head(n = 5)

# Plot creation
ggplot(genre_revenue_top5, aes(
  x = reorder(str_wrap(genre, 25), total_revenue),
  y = total_revenue
)) +
  geom_col(fill = "#6082B6") +

  geom_text(
    aes(label = paste0(round(total_revenue / 1e6, 1), "M")),
    hjust = -0.1,
    size = 4,
    fontface = "bold",
    color = "#1F1F1F"
  ) +

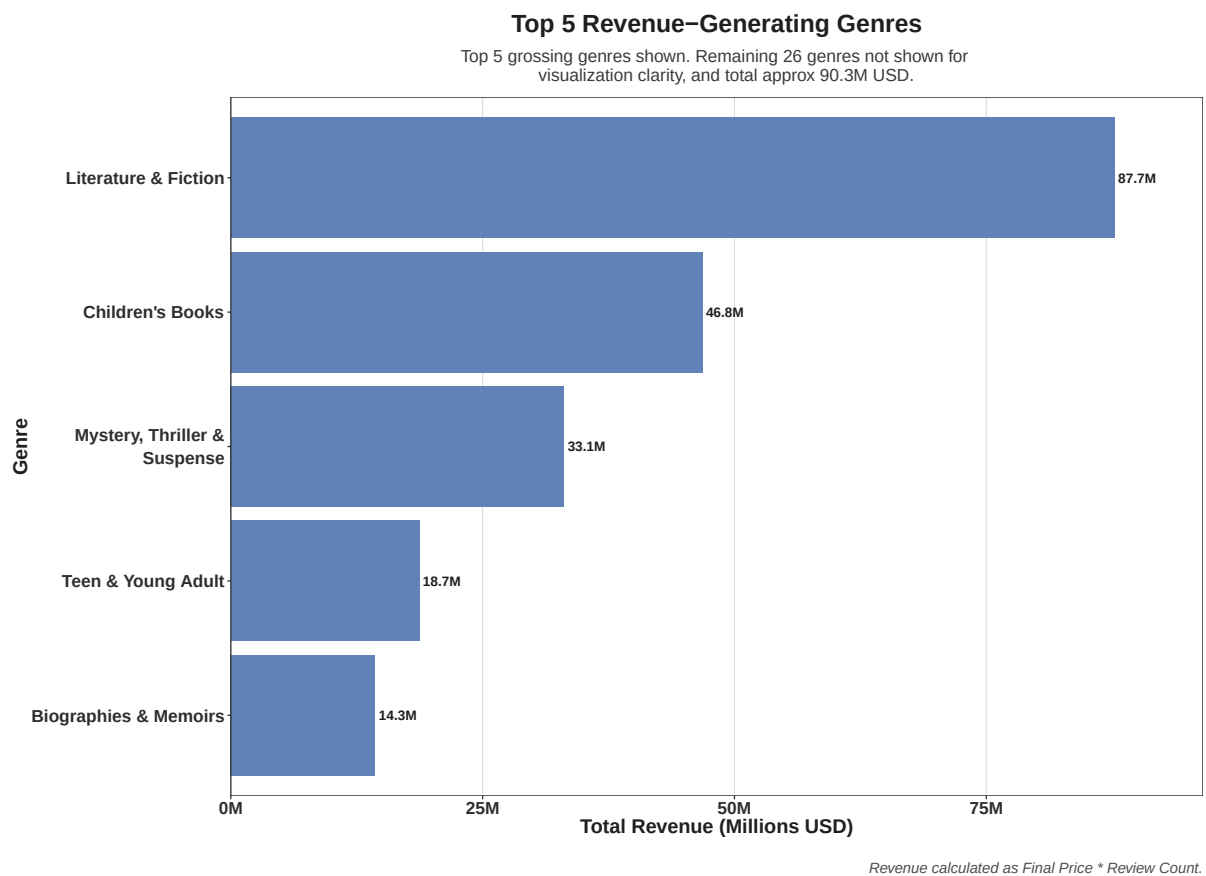
  coord_flip() +

  scale_y_continuous(
    labels = label_number(scale = 1e-6, suffix = "M"),
    expand = expansion(mult = c(0, 0.1))
  ) +

```

```
labs(
  title = "Top 5 Revenue-Generating Genres",
  subtitle = "Top 5 grossing genres shown. Remaining 26 genres not shown for
  visualization clarity, and total approx 90.3M USD.",
  x = "Genre",
  y = "Total Revenue (Millions USD)",
  caption = "Revenue calculated as Final Price * Review Count."
) +

theme_custom()
```



2. Highest Grossing Subgenres

```
Dataset <- Dataset |>
  mutate(revenue = final_price * reviews_count)

# Top 5 genres by revenue, excluding Literature & Fiction
top_genres <- Dataset |>
  filter(genre != "Literature & Fiction") |>
  group_by(genre) |>
  summarize(
```

```

    total_revenue = sum(revenue, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(total_revenue)) |>
  slice_head(n = 4) |>
  pull(genre)

# Top 5 subgenres within each top genre
genre_subgenre_revenue <- Dataset |>
  filter(
    genre %in% top_genres,
    genre != "Literature & Fiction"
  ) |>
  group_by(genre, subgenre) |>
  summarize(
    total_revenue = sum(revenue, na.rm = TRUE),
    .groups = "drop"
  ) |>
  group_by(genre) |>
  arrange(desc(total_revenue), .by_group = TRUE) |>
  slice_head(n = 5) |>
  mutate(rank = row_number()) |>
  ungroup()

# Pad each genre to 5 rows
genre_subgenre_padded <- genre_subgenre_revenue |>
  complete(
    genre = top_genres,
    rank = 1:5,
    fill = list(
      subgenre = NA_character_,
      total_revenue = 0
    )
  ) |>
  group_by(genre) |>
  arrange(rank, .by_group = TRUE) |>
  mutate(
    display_label = if_else(
      is.na(subgenre),
      "",
      str_wrap(subgenre, width = 18)
    ),
    plot_order = factor(rank, levels = rev(1:5))
  ) |>
  ungroup()

```

```

# Plot creation
make_subgenre_plot <- function(selected_genre) {

  plot_data <- genre_subgenre_padded |>
    filter(genre == selected_genre) |>
    arrange(rank)

```



```

label_lookup <- setNames(
  plot_data$display_label,
  as.character(plot_data$plot_order)
)

ggplot(plot_data, aes(
  x = plot_order,
  y = total_revenue,
  fill = total_revenue
)) +
  geom_col(width = 0.7) +

  geom_text(
    data = plot_data |> filter(total_revenue > 0),
    aes(
      label = label_number(
        scale = 1e-6,
        suffix = "M",
        accuracy = 0.1
      )(total_revenue)
    ),
    hjust = -0.1,
    size = 4,
    fontface = "bold",
    color = "#1F1F1F"
  ) +

  coord_flip(
    ylim = c(0, 27000000),
    clip = "off"
  ) +

  scale_x_discrete(
    labels = label_lookup
  ) +

  scale_y_continuous(
    labels = label_number(scale = 1e-6, suffix = "M"),
    breaks = seq(0, 25000000, by = 5000000),
    expand = expansion(mult = c(0, 0.05))
  ) +

  scale_fill_gradient(
    low = "#B6D0E2",
    high = "#6082B6"
  ) +

  labs(
    title = selected_genre,
    x = NULL,
    y = NULL
  ) +

```

```

    theme_custom() +
    theme(
      plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
      axis.text.y = element_text(size = 12, face = "bold"),
      axis.text.x = element_text(size = 12, face = "bold"),
      legend.position = "none",
      plot.margin = margin(8, 18, 8, 8)
    )
  }

plots <- lapply(top_genres, make_subgenre_plot)

combined_subgenre_plot <- wrap_plots(plots, ncol = 2)

x_label <- ggplot() +
  annotate(
    "text",
    x = 0.5,
    y = 0.5,
    label = "Revenue (Millions USD)",
    fontface = "bold",
    size = 6
  ) +
  theme_void()

y_label <- ggplot() +
  annotate(
    "text",
    x = 0.5,
    y = 0.5,
    label = "Subgenre",
    angle = 90,
    fontface = "bold",
    size = 6
  ) +
  theme_void()

axis_labeled_plot <-
  (y_label + combined_subgenre_plot +
   plot_layout(widths = c(0.02, 1))) /
  x_label +
  plot_layout(heights = c(1, 0.04))

final_layout <- axis_labeled_plot +
  plot_annotation(
    title = "Top Revenue-Generating Subgenres Within Top 5 Genres",
    subtitle = "Literature & Fiction genre was excluded due to its disproportionately
high revenue compared to other genres",
    caption = "Revenue calculated as Final Price * Review Count (used here as a
sales-demand proxy).",
    theme = theme(
      plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
      plot.subtitle = element_text(size = 14, hjust = 0.5),

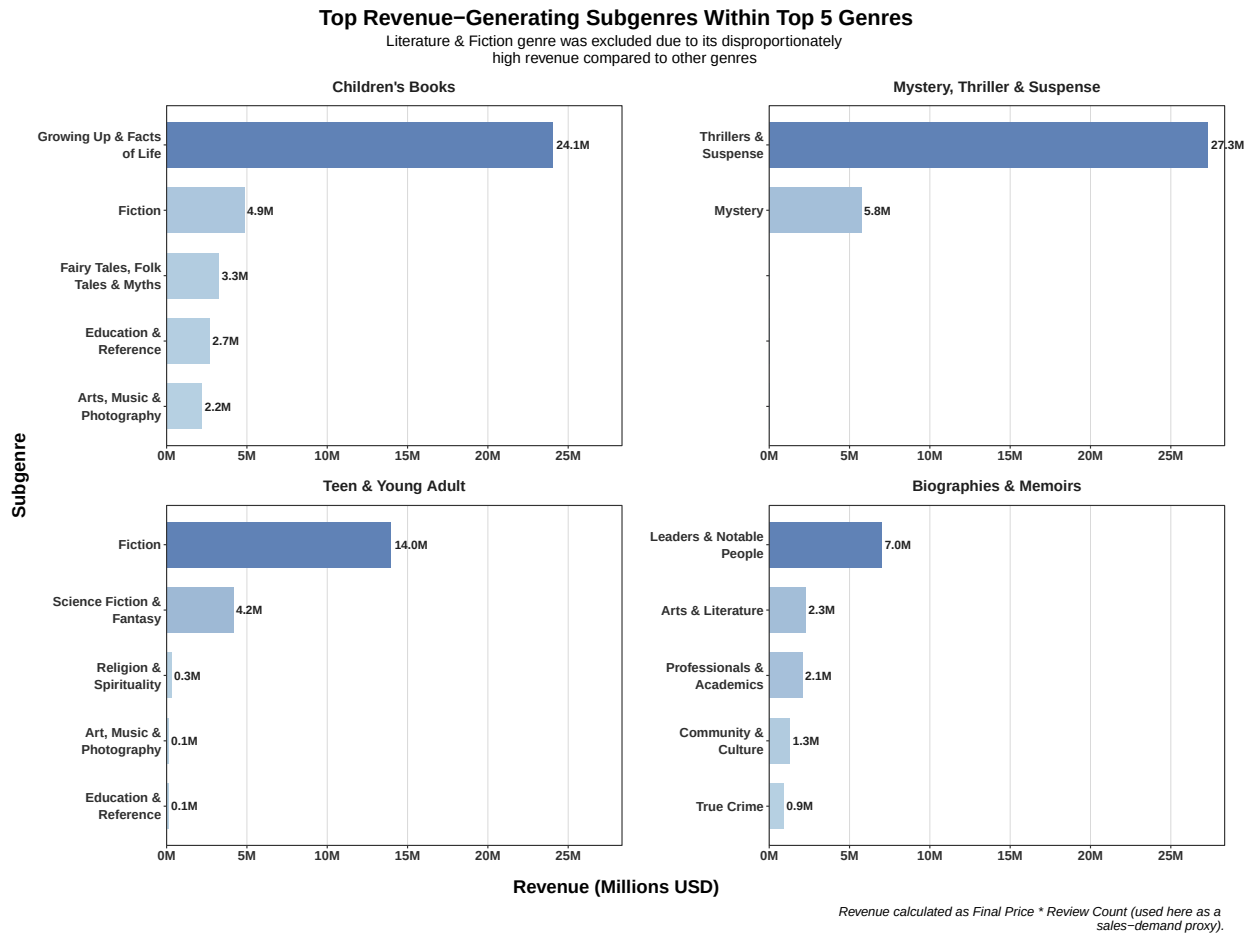
```

```

    plot.caption = element_text(size = 12, face = "italic", hjust = 1)
  )
)

final_layout

```



3. Priority Individual Books

```

# Define book series
selected_families <- c(
  "Harry Potter",
  "Lord of the Rings / The Hobbit",
  "Game of Thrones",
  "The Hunger Games",
  "Divergent",
  "Throne of Glass",
  "ACOTAR",
  "After",
  "Fifty Shades of Grey",
  "Blood and Ash",

```

```

"Outlander",
"Shadow and Bone",
"Amos Decker",
"The Stormlight Archive",
"Atlee Pine",
"Killing",
"The Heroes of Olympus",
"Aloysius Archer",
"Twilight Saga"
)

# Create prioritization based on revenue & engagement per series
series_priority <- Dataset |>
  filter(title_family %in% selected_families) |>
  group_by(title_family) |>
  summarize(
    revenue = sum(revenue, na.rm = TRUE),
    reviews_count = sum(reviews_count, na.rm = TRUE),
    number_of_titles = n_distinct(title_core),
    .groups = "drop"
  ) |>
  mutate(
    revenue_z = as.numeric(scale(revenue)),
    engagement_z = as.numeric(scale(reviews_count)),
    priority_score = 0.6 * revenue_z + 0.4 * engagement_z
  ) |>
  arrange(desc(priority_score))

# List only top ten series
top_series_priority <- series_priority |>
  slice_head(n = 10)

```

```

# Same as above but for individual titles
non_series_priority <- Dataset |>
  filter(!title_family %in% selected_families) |>
  group_by(title_core) |>
  summarize(
    revenue = sum(revenue, na.rm = TRUE),
    reviews_count = sum(reviews_count, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(
    revenue_z = as.numeric(scale(revenue)),
    engagement_z = as.numeric(scale(reviews_count)),
    priority_score = 0.6 * revenue_z + 0.4 * engagement_z
  ) |>
  arrange(desc(priority_score))

top_non_series_priority <- non_series_priority |>
  slice_head(n = 10)

```

```

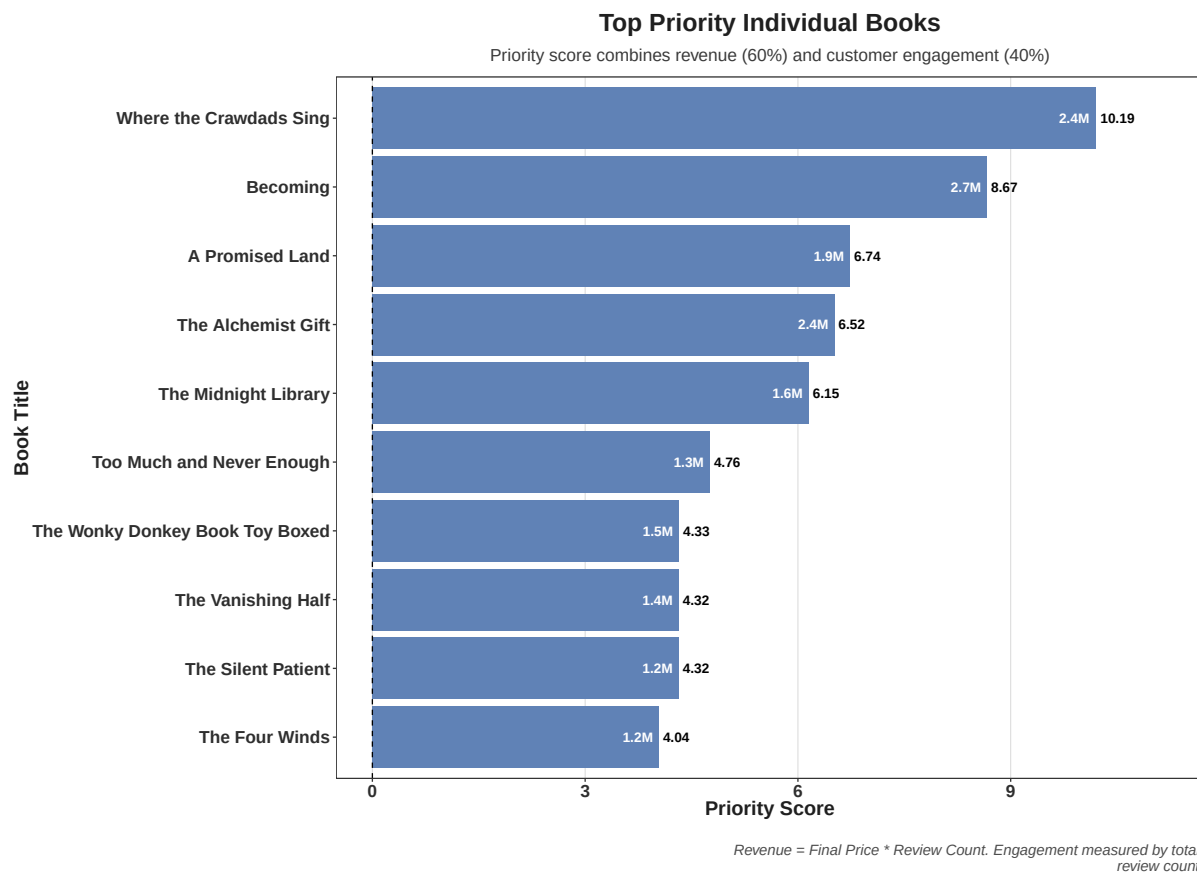
# Plot creation
ggplot(

```

```

top_non_series_priority,
aes(
  x = reorder(str_wrap(title_core, width = 35), priority_score),
  y = priority_score,
  fill = priority_score > 0
)
) +
geom_col() +
geom_text(
  aes(label = paste0(round(revenue / 1e6, 1), "M")),
  hjust = ifelse(top_non_series_priority$priority_score > 0, 1.2, -0.2),
  color = "white",
  size = 4,
  fontface = "bold"
) +
geom_text(
  aes(label = round(priority_score, 2)),
  hjust = ifelse(top_non_series_priority$priority_score > 0, -0.15, 1.15),
  size = 4,
  fontface = "bold"
) +
geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
coord_flip() +
scale_y_continuous(
  expand = expansion(mult = c(0.05, 0.15))
) +
scale_fill_manual(
  values = c("TRUE" = "#6082B6", "FALSE" = "#CC5801")
) +
labs(
  title = "Top Priority Individual Books",
  subtitle = "Priority score combines revenue (60%) and customer engagement (40%)",
  x = "Book Title",
  y = "Priority Score",
  caption = "Revenue = Final Price * Review Count. Engagement measured by total
  review count."
) +
theme_custom() +
guides(fill = "none")

```



4. High Priority & Grow Series Individual Titles

```
# Define book series
selected_families <- c(
  "Harry Potter",
  "Lord of the Rings / The Hobbit",
  "Game of Thrones",
  "The Hunger Games",
  "Divergent",
  "Throne of Glass",
  "ACOTAR",
  "After",
  "Fifty Shades of Grey",
  "Blood and Ash",
  "Outlander",
  "Shadow and Bone",
  "Amos Decker",
  "The Stormlight Archive",
  "Atlee Pine",
  "Killing",
  "The Heroes of Olympus",
  "Aloysius Archer",
```

```

"Twilight Saga"
)

# Revenue per series
family_title_revenue <- Dataset |>
  filter(title_family %in% selected_families) |>
  group_by(title_family, title_core) |>
  summarize(
    title_revenue = sum(revenue, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(title_family, desc(title_revenue))

# Plot creation
ggplot(series_priority, aes(
  x = engagement_z,
  y = revenue_z,
  label = title_family
)) +

  geom_point(
    size = 6,
    color = "#6082B6",
    alpha = 0.85
  ) +

  geom_hline(yintercept = 0, linetype = "solid", color = "#CC5801", linewidth = 1.3) +
  geom_vline(xintercept = 0, linetype = "solid", color = "#CC5801", linewidth = 1.3) +

  geom_text_repel(
    size = 5,
    fontface = "bold",
    color = "#1F1F1F",
    max.overlaps = Inf,
    box.padding = 0.5,
    point.padding = 0.4,
    min.segment.length = 0,
    segment.color = "#999999",
    segment.linewidth = 0.3
  ) +

  annotate("text", x = 0.55, y = 0.75, label = "High Priority", size = 6, fontface = "bold") +
  annotate("text", x = -0.75, y = 0.75, label = "Monetize", size = 6, fontface = "bold") +
  annotate("text", x = 0.55, y = -0.75, label = "Grow", size = 6, fontface = "bold") +
  annotate("text", x = -0.75, y = -0.75, label = "Low Priority", size = 6, fontface = "bold") +

  coord_cartesian(
    xlim = c(-1.05, 1.05),
    ylim = c(-1.05, 1.05),
    clip = "off"
  ) +

  labs(

```

```

title = "Strategic Prioritization of Book Series",
subtitle = "Prioritization matrix compares book series by standardized revenue
and engagement",
x = "Customer Engagement (Standardized Review Count)",
y = "Revenue (Standardized)",
caption = "Revenue = Final Price * Review Count. Engagement = total review count.
\nHarry Potter remains in the dataset but falls outside this zoomed view due to
dominant performance."
) +

theme_custom() +

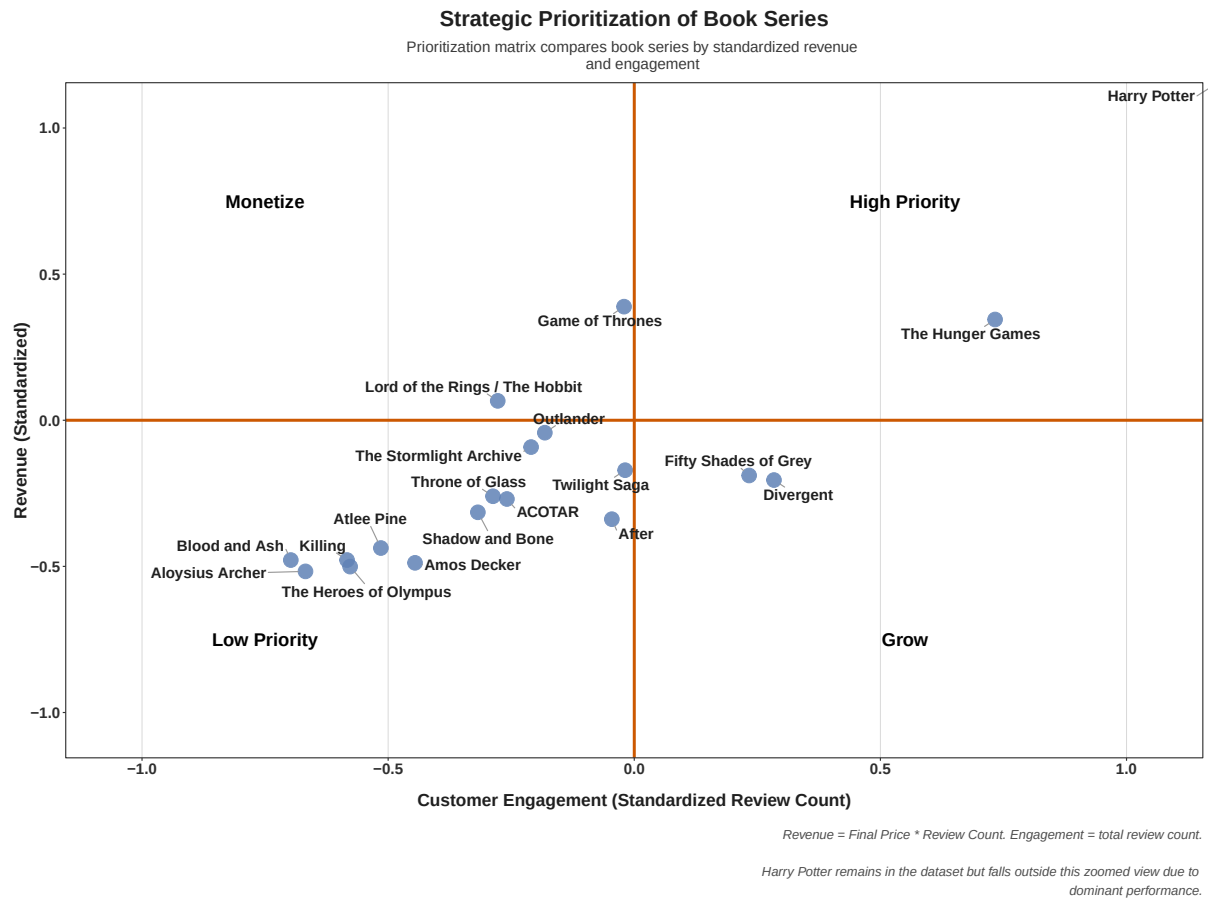
theme(
  plot.margin = margin(15, 40, 30, 15),
  axis.title.x = element_text(margin = margin(t = 15)),
  plot.caption = element_text(
    size = 12,
    face = "italic",
    hjust = 1,
    lineheight = 1.2,
    margin = margin(t = 18)
  )
)

```

```

## Warning in geom_text_repel(size = 5, fontface = "bold", color = "#1F1F1F", :
## Ignoring unknown parameters: 'segment.linewidth'

```

Additional data for series revenue

```
selected_families <- c(
  "Harry Potter",
  "Lord of the Rings / The Hobbit",
  "Game of Thrones",
  "The Hunger Games",
  "Divergent",
  "Throne of Glass",
  "ACOTAR",
  "After",
  "Fifty Shades of Grey",
  "Blood and Ash",
  "Outlander",
  "Shadow and Bone",
  "Amos Decker",
  "The Stormlight Archive",
  "Atlee Pine",
  "Killing",
  "The Heroes of Olympus",
  "Aloysius Archer",
  "Twilight Saga"
```

```

)

series_priority <- Dataset |>
  filter(title_family %in% selected_families) |>
  group_by(title_family) |>
  summarize(
    revenue = sum(revenue, na.rm = TRUE),
    reviews_count = sum(reviews_count, na.rm = TRUE),
    number_of_titles = n_distinct(title_core),
    .groups = "drop"
  ) |>
  mutate(
    revenue_z = as.numeric(scale(revenue)),
    engagement_z = as.numeric(scale(reviews_count)),
    priority_score = 0.6 * revenue_z + 0.4 * engagement_z
  ) |>
  arrange(desc(priority_score))

top_series_priority <- series_priority |>
  slice_head(n = 10)

```

```

ggplot(
  top_series_priority,
  aes(
    x = reorder(str_wrap(title_family, width = 30), priority_score),
    y = priority_score,
    fill = priority_score > 0
  )
) +
  geom_col() +

  # Revenue labels (inside bars)
  geom_text(
    aes(label = paste0(round(revenue / 1e6, 1), "M")),
    hjust = ifelse(top_series_priority$priority_score > 0, 1.2, -0.2),
    color = "white",
    size = 4,
    fontface = "bold"
  ) +

  # Priority score labels (outside bars)
  geom_text(
    aes(label = round(priority_score, 2)),
    hjust = ifelse(top_series_priority$priority_score > 0, -0.15, 1.15),
    size = 4,
    fontface = "bold"
  ) +

  # Reference line at 0
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +

  coord_flip() +

```

```

scale_y_continuous(
  expand = expansion(mult = c(0.05, 0.15))
) +

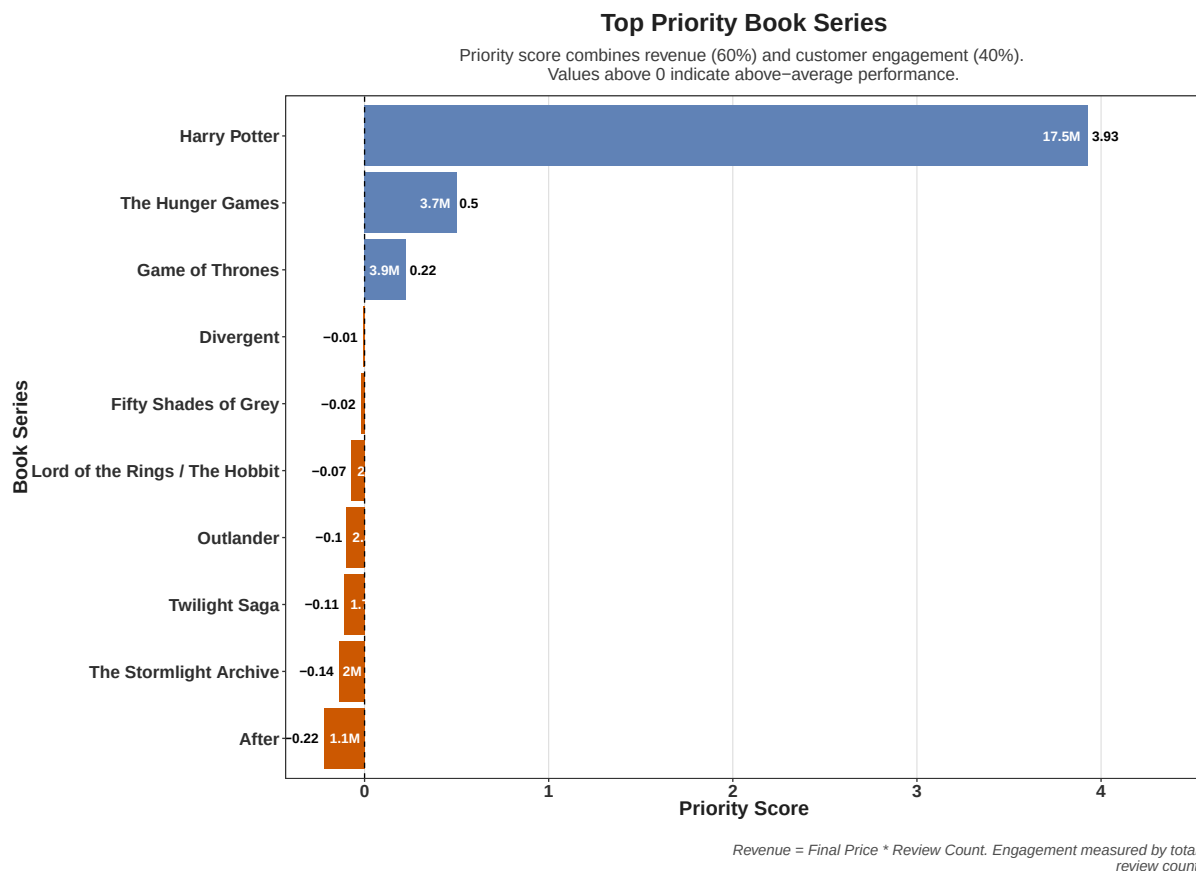
# Color coding positive vs negative
scale_fill_manual(
  values = c("TRUE" = "#6082B6", "FALSE" = "#CC5801")
) +

labs(
  title = "Top Priority Book Series",
  subtitle = "Priority score combines revenue (60%) and customer engagement (40%).  
Values above 0 indicate above-average performance.",
  x = "Book Series",
  y = "Priority Score",
  caption = "Revenue = Final Price * Review Count. Engagement measured by total  
review count."
) +

theme_custom() +

guides(fill = "none")

```



5. Top Grossing Titles within a Series

```
# Identify priority series & their revenue
priority_families <- series_priority |>
  filter(engagement_z > 0) |>
  pull(title_family)

family_title_revenue_priority <- Dataset |>
  filter(title_family %in% priority_families) |>
  filter(!(title_family == "Harry Potter" & title_core == "Harry Potter")) |>
  group_by(title_family, title_core) |>
  summarize(
    title_revenue = sum(revenue, na.rm = TRUE),
    .groups = "drop"
  ) |>
  group_by(title_family) |>
  slice_max(title_revenue, n = 5, with_ties = FALSE) |>
  arrange(title_family, desc(title_revenue)) |>
  mutate(rank = row_number()) |>
  ungroup()

family_title_revenue_padded <- family_title_revenue_priority |>
  complete(
    title_family,
    rank = 1:5,
    fill = list(
      title_core = "",
      title_revenue = 0
    )
  ) |>
  group_by(title_family) |>
  arrange(rank, .by_group = TRUE) |>
  mutate(
    display_label = if_else(
      title_core == "",
      "",
      str_wrap(title_core, width = 18)
    ),
    plot_order = factor(rank, levels = rev(1:5))
  ) |>
  ungroup()

make_family_plot <- function(selected_family) {

  plot_data <- family_title_revenue_padded |>
    filter(title_family == selected_family) |>
    arrange(rank)

# Plot creation starts here
  ggplot(plot_data, aes(
    x = plot_order,
    y = title_revenue,
    fill = title_revenue
```

```

)) +
  geom_col(width = 0.7) +
  geom_text(
    data = plot_data |> filter(title_revenue > 0),
    aes(
      label = label_number(
        scale = 1e-6,
        suffix = "M",
        accuracy = 0.1
      )(title_revenue)
    ),
    hjust = -0.1,
    size = 4,
    fontface = "bold",
    color = "#1F1F1F"
  ) +
  coord_flip() +
  scale_x_discrete(labels = rev(plot_data$display_label)) +
  scale_y_continuous(
    limits = c(0, 4000000),
    labels = label_number(scale = 1e-6, suffix = "M"),
    expand = expansion(mult = c(0, 0.05))
  ) +
  scale_fill_gradient(
    low = "#B6D0E2",
    high = "#6082B6"
  ) +
  labs(
    title = selected_family,
    x = NULL,
    y = NULL
  ) +
  theme_custom() +
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.text.y = element_text(size = 12, face = "bold"),
    axis.text.x = element_text(size = 12, face = "bold"),
    legend.position = "none",
    plot.margin = margin(8, 18, 8, 8)
  )
}

family_plots <- lapply(priority_families, make_family_plot)

combined_family_plot <- wrap_plots(family_plots, ncol = 2)

x_label <- ggplot() +
  annotate(
    "text",
    x = 0.5,
    y = 0.5,
    label = "Revenue (Millions USD)",
    fontface = "bold",

```

```

    size = 6
  ) +
  theme_void()

y_label <- ggplot() +
  annotate(
    "text",
    x = 0.5,
    y = 0.5,
    label = "Individual Title",
    angle = 90,
    fontface = "bold",
    size = 6
  ) +
  theme_void()

axis_labeled_plot <-
  (y_label + combined_family_plot +
    plot_layout(widths = c(0.02, 1))) /
  x_label +
  plot_layout(heights = c(1, 0.04))

final_family_priority_plot <- axis_labeled_plot +
  plot_annotation(
    title = "Revenue Contribution of Titles Within High-Opportunity Book Series",
    subtitle = "View of High Priority and Grow series; these have above-average engagement.",
    caption = "Revenue calculated as Final Price * Review Count. \n\nThe aggregate 'Harry Potter'
    title entry was excluded to improve visibility of individual title contributions within the
    series.",
    theme = theme(
      plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
      plot.subtitle = element_text(size = 14, hjust = 0.5),
      plot.caption = element_text(size = 12, face = "italic", hjust = 1)
    )
  )

final_family_priority_plot

```

```

## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_col()').

```

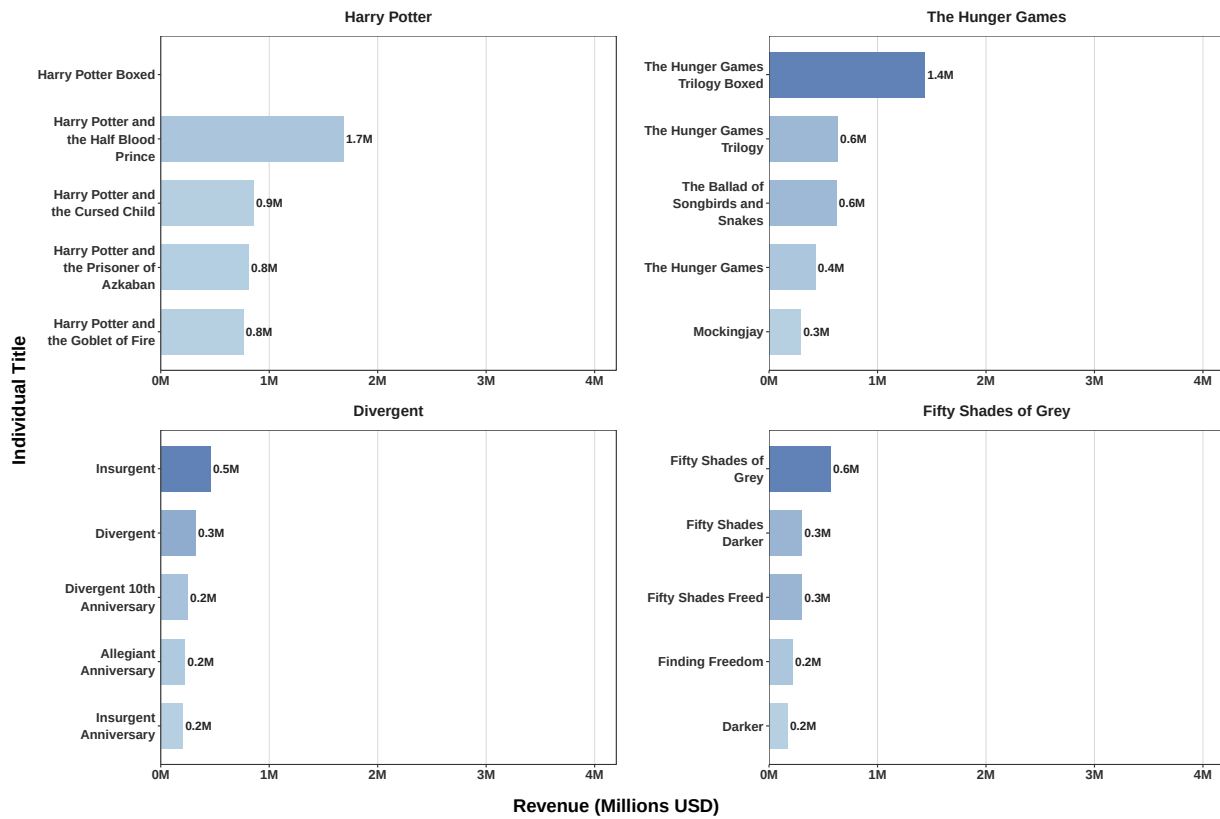
```

## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_text()').

```

Revenue Contribution of Titles Within High-Opportunity Book Series

View of High Priority and Grow series; these have above-average engagement.



Revenue calculated as Final Price * Review Count.
The aggregate 'Harry Potter' title entry was excluded to improve visibility of individual title contributions within the series.